

End-to-End UAV Security is a Two-Layer Problem: A Cross-Layer Benchmark Spanning the Network and Navigation Attack Surfaces of Autonomous Drones

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Abstract—Security research for unmanned aerial vehicles (UAVs) is split down the middle. On one side, intrusion-detection datasets model attacks *on the wire*—swarm-mesh routing, the ground-control link, the on-board bus—and stop at the packet. On the other, autonomy-security work models attacks *on the flight*—GNSS spoofing, jamming, sensor perturbation—and takes the network for granted. No existing benchmark lets a researcher ask the question that actually decides whether a drone completes its mission: *when a network attack slips past the detector, what does it do to the aircraft?* We present the first end-to-end UAV-security benchmark that spans both layers. It unifies six datasets—four established network-layer corpora and two navigation-layer sources—under a single two-layer schema whose central object is not a packet or a flight but a *cross-layer pairing*: a network attack window joined to the navigation degradation it induces, tagged with an honesty flag recording whether that coupling was measured on one platform, derived through a physical model, or aligned in simulation. We contribute the schema, six validated ingestion adapters, a detector-operating-curve harness, and a reference cross-layer pairing pipeline, released as an open artifact. Across the corpus we show that the network and navigation layers, studied jointly, expose failure modes invisible to either alone—most sharply, that a time-delay forwarding attack tuned to evade a state-of-the-art detector still drives measurable navigation staleness. The benchmark is designed as the experimental substrate for composed, cross-layer security guarantees.

Index Terms—UAV security, intrusion detection, GNSS spoofing, benchmark, cross-layer, cyber-physical systems, electronic warfare.

I. INTRODUCTION

A modern autonomous UAV is a stack of trust boundaries. Commands descend from a ground station over a MAVLink command-and-control (C2) link; peer aircraft exchange state over a multi-hop swarm mesh; motor controllers and sensors

talk over an on-board UAVCAN/DroneCAN bus; and a navigation pipeline fuses GNSS, inertial, and visual measurements into the state estimate that closes the flight control loop. An adversary can enter at any of these boundaries, and the security literature has produced strong, specialised defences at each. What it has *not* produced is a way to reason across them.

This gap is not academic. Electronic-warfare (EW) environments routinely defeat drones not by breaking cryptography but by degrading the physics: a jammer raises the jamming-to-signal ratio (J/S) until the GNSS solution collapses and the mission fails [1]. A forwarding attack in the mesh does not exfiltrate data; it delays a correction until the aircraft is acting on stale state. In every case the *network* event and the *navigation* outcome are two ends of one causal chain—and yet no benchmark represents that chain. Network intrusion-detection datasets for UAVs [2]–[4] label the packet and stop; UAV autonomy-security work [5], [6] measures the flight and assumes the network. A defender who wants to know whether a detector tuned to a given false-positive budget actually keeps the aircraft flying has nowhere to look.

We close this gap with a benchmark whose unit of analysis is the cross-layer pairing itself.

To see why a single-layer benchmark is not enough, consider what a reviewer of a UAV intrusion detector can and cannot currently check. They can verify that the detector flags a delayed forwarding event or a spoofed GNSS message with high recall. They *cannot* verify the claim that actually matters to an operator—that a detector tuned to a given false-alarm budget keeps aircraft completing their missions—because no dataset records both the network event and the flight it degrades. Our benchmark makes that claim checkable by pairing, for the first time, the two halves of the causal chain in a single schema and a single artifact.

Contributions

- **A two-layer schema (§IV).** We define a single schema that represents any observation from any of the six datasets as a typed *Event*, groups events into labelled

Preprint prepared for IEEE SaTML. Artifact and data: <https://www.kaggle.com/datasets/rogernickanaedevha/uavs-network-and-navigation-end-to-end-security-data> (DOI 10.34740/kaggle/dsv/18346203). Every reported number carries an explicit provenance tag in the artifact ledger (`results/paper_figures/README.md`): real-corpus, simulation-derived, or a stated modeling parameter.

AttackWindows, and—the novel part—joins a network window to an autonomy window as a *CrossLayerPairing* carrying the detector operating point, the residual undetected budget, the induced perturbation, and the mission outcome. A per-record *pairing basis* flag (`measured_same_platform / physics_model / synthetic_alignment`) makes the strength of every coupling explicit and auditable.

- **Six validated ingestion adapters (§V).** We ingest four established network-layer corpora (UAVIDS-2025, UAV-CAS, the UAV ATTACK DATASET, and the HCRL UAVCAN bus dataset) and two navigation-layer sources (our physics-informed UAV-EW-BENCH-2026 and time-window-graph forwarding traces from a DATA-MUT-style detector), each mapped losslessly into the schema and validated against it.
- **A detector-operating-curve harness (§VI).** We make the network detector’s operating point a swept parameter and measure detection recall / false-positive rate as a function of it, producing the network-side input that a composed guarantee consumes.
- **A reference cross-layer pipeline and open artifact (§VII).** We release the schema, adapters, harness, and a proof-of-concept pairing pipeline as a reproducible artifact, and use it to demonstrate that an evasion-tuned network attack still induces measurable navigation degradation—the empirical hook for cross-layer defence.

We deliberately scope this paper to the *benchmark and its measurements*; the composed formal guarantee it enables is developed in a companion paper and previewed in §VIII.

II. BACKGROUND AND PRELIMINARIES

This section fixes terminology for readers who review security or machine-learning work but do not specialise in drones. Table I lists every acronym.

A. Anatomy of an autonomous UAV under attack

An unmanned aerial vehicle (UAV, or drone) flies by continuously estimating its own state—position, velocity, attitude—and steering to keep that estimate on a planned route. The estimate is produced by fusing several sensors, chiefly a satellite-navigation (GNSS, e.g. GPS) receiver and an inertial measurement unit (IMU), in an onboard filter; on the PX4 autopilot used by several datasets here, that filter is an extended Kalman filter (EKF). Around this single aircraft sit three communication surfaces an attacker can reach: a multi-hop *swarm mesh* over which peer drones relay one another’s traffic; a *command-and-control (C2) link* (typically the MAVLink protocol) to a ground control station; and an *intra-vehicle bus* (UAVCAN/DroneCAN) connecting the flight controller to motor controllers and sensors. We call these three the *network layer*. The GNSS receiver, the sensor fusion, the controller, and the mission outcome we call the *autonomy layer*. The thesis of this paper is that security must be evaluated across both layers at once, because the attacks that matter in

TABLE I: Acronyms, in order of first use.

UAV	unmanned aerial vehicle (drone)
IDS	intrusion detection system
GNSS / GPS	Global Navigation Satellite System / Global Positioning System
IMU	inertial measurement unit
EKF	extended Kalman filter (onboard sensor-fusion estimator)
C2	command and control (ground-to-aircraft link)
MAVLink	Micro Air Vehicle Link, standard UAV C2 protocol
UAVCAN	UAV Controller Area Network (intra-vehicle bus; a.k.a. DroneCAN)
ESC	electronic speed controller (motor driver on the bus)
DTN	delay-tolerant network (store-and-forward mesh)
TWiG	time-window graph (DATAMut’s forwarding-schedule model)
RF	radio frequency
EW	electronic warfare
J/S	jamming-to-signal power ratio, in dB
MCR	Mission Completion Rate
FPR	false-positive rate
DO-326A	RTCA/EUROCAE airworthiness-security process standard
DoS / DDoS	(distributed) denial of service
NS-3	a widely used discrete-event network simulator
AERPAW	Aerial Experimentation and Research Platform for Advanced Wireless

the field—jamming, spoofing, forwarding delay—are network events whose damage is an autonomy outcome.

B. The two attack surfaces

We partition the UAV attack surface into two *layers*. The **network** layer covers everything between nodes: the swarm mesh (multi-hop routing among aircraft), the C2 link (ground station to UAV, typically MAVLink), and the intra-UAV bus (UAVCAN/DroneCAN among flight controller, ESCs, and sensors). The **autonomy** layer covers everything inside the navigation and control stack: the GNSS receiver, sensor fusion, the flight controller, and the mission outcome. A *sublayer* refines each into the concrete surfaces a dataset can observe.

C. Threat model

We assume an adversary who can (i) inject, drop, delay, or forge traffic on one or more network sublayers, and (ii) manipulate the RF environment of the GNSS receiver (spoofing or jamming). We assume a network-layer *detector* operating at a tunable point θ that trades detection recall against false-positive rate. The security question the benchmark is built to ask is *cross-layer*: given a detector at θ , and an attacker who tunes its behaviour to stay just under θ , what navigation degradation survives, and does the mission still complete? This paper measures the ingredients of that question; it does not yet bound the answer (companion paper). Fig. 1 places the attack surfaces our six datasets cover on the causal chain from wire to flight.

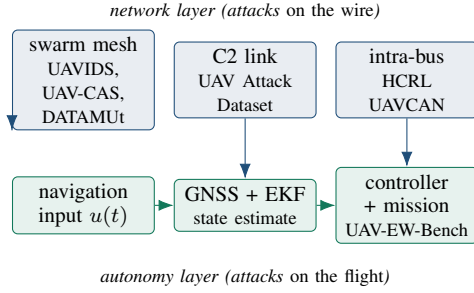


Fig. 1: The attack surfaces the six datasets cover, arranged on the causal chain. Network-layer events (top) propagate downward into the autonomy layer (bottom) through the navigation input; the benchmark’s contribution is to represent that downward arrow explicitly rather than assume it.

D. Mission Completion Rate and the regulatory floor

We adopt *Mission Completion Rate* (MCR)—the fraction of flights that reach their objective under attack—as the operational metric, and take the $\text{MCR} \geq 0.90$ line as an operational reference consistent with the airworthiness-security posture of DO-326A/ED-202A [7]. Emerging AI-in-aviation assurance guidance [8] motivates treating a certified MCR floor as a first-class safety property, which the companion paper takes up.

E. Problem formulation: what a cross-layer benchmark must represent

Let a network attack occupy a time window $W_n = [t_0, t_1]$ on some network sublayer, and let the induced navigation degradation occupy a window W_a on the autonomy layer. Existing datasets provide either W_n (with packet labels) or W_a (with flight outcomes) but never the *pair* (W_n, W_a) together with the mechanism linking them. We formalise the missing object as a tuple

$$P = (W_n, W_a, \theta, \Delta(\theta), \delta, \text{MCR}), \quad (1)$$

where θ is the detector operating point, $\Delta(\theta)$ is the worst-case residual effect of an attacker that evades the detector at θ , δ is the resulting perturbation budget on the navigation input, and MCR is the measured (or, in the companion paper, certified) outcome. A benchmark that represents P —not just W_n or W_a —is the contribution of this paper. Crucially, the honesty of P depends on *how* W_n and W_a were linked, which we record explicitly (§IV).

III. RELATED WORK

We survey from the general to the specific: first the state of UAV intrusion-detection datasets, then the network sublayers each covers, then the navigation-security work on the other side of the gap, and finally the cross-layer efforts closest to ours.

a) UAV IDS datasets and surveys: Recent taxonomies of AI-enhanced UAV intrusion detection [9] document a proliferation of datasets and a recurring weakness: general-purpose network corpora lack UAV-specific protocol semantics, and UAV-specific corpora each cover a single sublayer. [9] reports that models trained on generic benchmarks misclassify a large fraction of GPS-spoofing events on real autopilot logs—direct evidence that the network and navigation layers cannot be treated as one undifferentiated feature space. FANET security surveys [10], [11] catalogue routing attacks (blackhole, wormhole, Sybil, flooding) but stop at network effect.

b) Network-layer datasets by sublayer: On the swarm mesh, UAVIDS-2025 [2], [12] provides 122,171 NS-3 flow records across four attack classes, and UAV-CAS [3] provides 99,492 flows from a digital twin calibrated against the AER-PAW testbed, with collaborative multi-attack compositions. On the C2 link, the UAV ATTACK DATASET [5], [13] records real PX4 flights under GPS spoofing/jamming and MAVLink ping-DoS. On the intra-UAV bus, the HCRL UAVCAN dataset [4] captures DroneCAN flooding, fuzzing, and replay. Each is authoritative for its sublayer; none reaches the navigation outcome.

c) Forwarding / time-delay detection: DATAMUt [14] detects time-delay attacks in multi-hop UAV networks using a deterministic time-window-graph model, with holistic cross-layer variants [15] pointing toward—but not delivering—joint network–navigation evaluation. We adopt a DATAMUt-style detector as the network-side exemplar precisely because its operating point is a clean, sweepable threshold.

d) Navigation-layer security: On the other side of the gap, GNSS spoofing/jamming has its own mature line—capture-via-spoofing [6], [16], the TEXBAT test battery [17], and resilient GPS-denied navigation [1], which quantifies how jamming drives position error and, ultimately, mission loss. Learning-based spoofing detectors now reach high accuracy on curated GNSS datasets [18], [19], but they are trained and evaluated on receiver-level features in isolation from any mission outcome. This work measures the flight but abstracts away the network path by which an attack arrives.

e) Why dataset construction is itself a research contribution: The intrusion-detection community has learned, sometimes painfully, that a benchmark’s *construction* determines the validity of every result built on it. Careful audits of the widely used CIC-IDS2017 corpus uncovered labelling errors, flawed flow construction, and unrealistic temporal separation between attack and benign traffic that materially inflate reported performance [20], [21]. The lesson is directly relevant to a *cross-layer* benchmark, where a new failure mode appears: the temptation to assert that a given network event caused a given flight degradation when the two were never observed together. Our schema answers this with a mandatory per-record honesty flag (§IV) that records exactly how each network–autonomy pairing was established, so a reviewer can filter the benchmark by evidence strength rather than trusting an unaudited join. We treat provenance discipline—not merely scale—as a first-class design requirement.

f) *Positioning against the closest prior art*: The nearest neighbour to our work in ambition is UAV-CAS, a large AERPAW-calibrated swarm dataset that is, notably, the first to include *collaborative* multi-attack compositions and to calibrate a simulator against real testbed measurements [3]. It is an excellent network-layer benchmark, and we ingest it as one of our six sources. But it remains within the network layer: its authors explicitly list GNSS spoofing and jamming—the autonomy layer—as future work. Our contribution is orthogonal and complementary: rather than deepen one layer, we *span* both and make the coupling between them the unit of analysis. Where UAV-CAS asks “how well can we detect a collaborative mesh attack?”, we ask “and what does that attack, once detected or missed, do to the flight?”—a question that requires the autonomy anchor and the pairing record that UAV-CAS, by design, does not provide.

g) *The cross-layer gap*: To our knowledge no released benchmark represents the pairing P of (1). The closest cross-layer detection framework [15] argues for holistic evaluation but does not ship a dataset spanning both layers with attack-to-degradation pairings; single-layer corpora, however excellent within their layer, cannot by construction. This is the gap we fill: not a larger dataset for one layer, but the first dataset whose records *are* the cross-layer couplings.

IV. THE TWO-LAYER SCHEMA

The schema has three record kinds, designed so the cross-layer pairing is a first-class object rather than an implicit join.

Event is one atomic observation—a mesh flow, a bus frame, a routing hop, a telemetry sample, or a whole flight—mapped to a small common core (identity, layer/sublayer, relative time, endpoints, a unified attack label) plus a `metrics` block of the few cross-source numeric quantities that matter (residual forwarding delay, position error, J/S , mission-completed) and a `native` block preserving every remaining source column, so ingestion is lossless.

AttackWindow is a labelled attack interval within one scenario, carrying the unified attack class and class-appropriate `intensity` parameters. It is the unit of cross-layer alignment.

CrossLayerPairing realises the tuple P of (1): it joins a network window to an autonomy window and records θ , the residual budget $\Delta(\theta)$, the perturbation δ with a *named, versioned* mapping, the certificate hook, and the empirical MCR. Its `pairing_basis` field is the schema’s conscience: `measured_same_platform` (both windows recorded on one testbed—the strongest evidence), `physics_model` (coupled through a stated physical model), or `synthetic_alignment` (aligned by construction). A reviewer can filter the benchmark by evidence strength; a claim can never silently upgrade its own provenance.

The design choices behind these three kinds are deliberate and worth making explicit, since they are what let six heterogeneous sources coexist without information loss (Table II). Time is always *relative* to each source’s own capture start, because the sources share no global clock; cross-layer

TABLE II: Top: the schema’s three record kinds and their load-bearing fields. Bottom: the unified attack taxonomy the adapters map onto, with the source-native label retained alongside. Full JSON Schema in the artifact.

Record kinds	
Event	atomic observation; core identity + layer/sublayer + relative time + metrics + lossless native
AttackWindow	labelled attack interval; unified class + intensity; the unit of alignment
CrossLayerPairing	(network, autonomy) window join + θ, Δ, δ , certificate, MCR, and pairing_basis
Unified attack taxonomy (label_native retained)	
network / mesh	blackhole, wormhole, Sybil, flooding, time-delay, collaborative
network / C2	GPS-spoof, GPS-jam, ping-DoS
network / bus	flooding, fuzzy, replay
autonomy / GNSS	spoofing, jamming (graded by J/S)

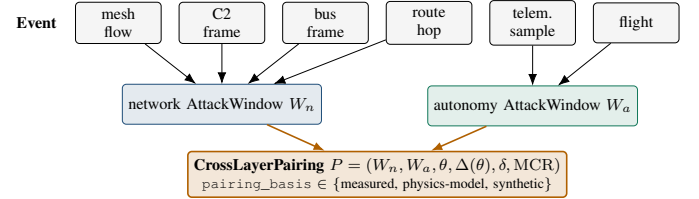


Fig. 2: The schema in one picture. Heterogeneous events from six datasets fold into typed network and autonomy *AttackWindows*; a *CrossLayerPairing* joins one of each and records the detector operating point, the residual budget, the induced perturbation, and the mission outcome, tagged by how the coupling was established.

alignment happens at the window level, not the event level, so a mismatch in sampling rate between a 50 Hz telemetry stream and a per-flow network record never forces a lossy resample. Attack labels are *unified* into a common taxonomy (Table II, lower half) while the verbatim source label is retained in `label_native`, so a model can train on the unified class and an auditor can still recover exactly what the source called it. And every unmapped column is preserved in `native`, which is what makes the ingestion provably lossless rather than merely convenient.

V. THE CORPUS AND INGESTION ADAPTERS

Table III summarises the six datasets and the layer each anchors. Every adapter emits schema-valid `events.jsonl` and `windows.jsonl`; unmapped source columns are preserved in `native`. We do not re-host raw third-party data; the artifact ships integration scripts and unified labels, and each source is fetched from its origin.

TABLE III: The six-dataset corpus. Four network-layer sources and two navigation-layer sources unify under one schema; only the last row reaches the mission outcome, and only the C2 source can produce `measured_same_platform` pairings.

Dataset	Real/Sim	Layer · sublayer	Event kind	Unique role in the end-to-end benchmark
UAVIDS-2025 [2]	Sim (NS-3.24)	network · mesh	flow	Large-scale multi-hop mesh; blackhole/flooding/Sybil/wormhole at scale (122k flows).
UAV-CAS [3]	Sim (digital twin)	network · mesh	flow	AERPAAW-calibrated swarm; collaborative multi-attack compositions (99k flows, 1024 configs).
UAV ATTACK DATASET [13]	Real testbed	network · C2 (MAVLink)	telemetry	Real PX4 GPS spoof/jam + ping-DoS on the <i>same</i> airframe family we model—the measured bridge.
HCRL UAVCAN [4]	Real testbed	network · intra-bus	frame	On-board UAVCAN/DroneCAN flooding/fuzzy/replay—a third attack surface below the C2 link.
DATAMUT traces [14]	Sim (event)	network · mesh (DTN)	hop	Time-window-graph forwarding detector with a <i>sweepable</i> operating point θ .
UAV-EW-BENCH-2026 [22]	Sim (physics-informed)	autonomy · GNSS+ctrl	flight	The navigation anchor: full 0–40 dB J/S sweep with per-flight MCR outcomes.

A. What each adapter does

An adapter’s job is lossless normalization: map a source’s native records onto the schema’s common fields and preserve everything else in a `native` block so no information is discarded. Three adapters warrant specifics because their sources are irregular. The UAV ATTACK DATASET adapter parses PX4 flight logs, computes horizontal position error against a benign reference (or, absent one, the flight’s own pre-attack median), and emits the `measured_same_platform` pairings that ground the interface mapping. The HCRL UAVCAN adapter parses candump-style bus frames; because the per-scenario attack labels are not machine-readable in the release, we *derive* each scenario’s class from the frames themselves—burst rate relative to the normal bus rate, the fraction of distinct payloads, and per-frame byte entropy cleanly separate a flooding burst (one payload hammered above bus rate) from a fuzzy burst (near-random payloads) from a replay burst (a small set of captured payloads reused). Table V reports the result; it is confirmed against the dataset’s technical report before camera-ready. The UAV-CAS adapter parses the calibrated digital twin’s flow statistics, including the stringified list columns that carry per-packet timing; it is verified against CSVs produced by the original authors’ own generator, and re-ingests the released statistics file verbatim as that file becomes locally reachable.

VI. THE DETECTOR-OPERATING-CURVE HARNESS

A cross-layer guarantee needs the network detector’s behaviour *as a function of its operating point*. For the DATAMUT-style detector the operating point is a deviation threshold ε (in seconds): a forwarding hop is flagged when its residual delay exceeds ε or its next hop deviates from the routing

TABLE IV: Real ingested corpus. Counts are from the schema-validated adapter outputs (`results/ingest_stats.csv`); every record validates against the schema of §IV. All values real-corpus except where a source’s release file was not yet locally reachable.

Source (real data ingested)	Events	Windows
UAV-EW-BENCH-2026	93,600	468
UAV ATTACK DATASET (live sample)	610	0
UAVIDS-2025 (shard 0)	122,171	4
HCRL UAVCAN (10 scenarios)	200,000	34
UAV-CAS (stat)	adapter ready; file pending	
DATAMUT traces	generated per sweep (15,392 hops)	

oracle. We expose ε as a swept parameter and, for each value, measure detection recall and false-positive rate over multiple seeds, scenarios, and two attacker delay regimes.

The resulting operating curve (Fig. 4) is the network-side input a composed guarantee consumes: it tells us, for any target false-positive budget, how large an undetected per-hop delay an evading attacker may inject—the residual budget $\Delta(\theta)$ of (1). The knee in the curve is structural: an evading attacker’s per-hop delay is additive up to the inter-node contact-window slack, beyond which a missed contact window forces a full-period penalty. This piecewise structure is what makes the cross-layer story sharp rather than smooth.

The knee, and the floor it descends to, are not simulation artifacts but a structural property of any store-and-forward schedule, and they recur across all three network topologies in our harness. Fig. 5 overlays the grid, ring, and double-

TABLE V: Frame-derived attack classes for the HCRL UAVCAN scenarios. Classes are inferred from burst rate, payload-distinct share, and byte entropy; scenarios that exhibit more than one burst shape map to `collaborative`.

Scenario	Frames	Bursts	Frame-derived class(es)
type1	207,858	3	flooding
type2	134,170	3	replay
type3	197,479	3	fuzzy
type4	133,374	3	fuzzy
type5	180,608	3	fuzzy
type6	241,321	4	fuzzy
type7	234,162	4	replay, fuzzy
type8	265,800	4	fuzzy
type9	230,378	4	replay, fuzzy
type10	207,380	3	replay, fuzzy

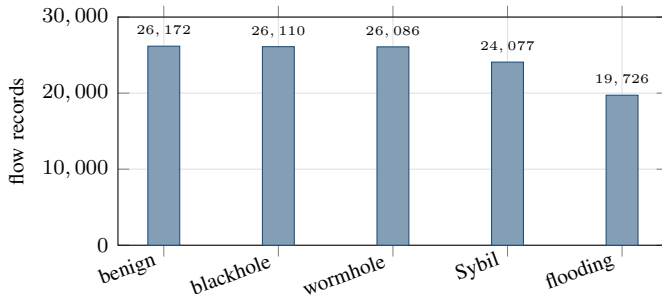


Fig. 3: Class distribution of the ingested UAVIDS-2025 swarm-mesh corpus (shard 0, 122,171 real flow records). The four attack classes are close to balanced against benign traffic—an intentional design property of the source that makes it a clean substrate for the network side of a pairing.

ring topologies. Two observations transfer directly to the companion guarantee. First, the floor height differs by topology because it equals the probability that a malicious delay overshoots the contact-window slack and self-exposes—higher in the medium delay regime ($U[1, 10]$ s) than in the low one ($U[1, 7]$ s), because a wider delay draw is likelier to overshoot. Second, no topology’s recall keeps falling to zero: past the knee the detector still catches the self-exposing fraction, which is precisely why an evading attacker’s undetected budget *saturates* rather than growing without bound. This is the empirical basis for the residual-budget cap that the companion paper turns into a certified floor.

VII. CROSS-LAYER MEASUREMENTS

A. The navigation anchor

The autonomy layer supplies the outcome the whole benchmark is built to reach. Fig. 6 shows MCR versus J/S for four defence configurations on UAV-EW-BENCH-2026. The undefended baseline collapses below the 0.90 floor by $J/S \approx 8$ dB, while the strongest configuration holds the floor to $J/S \approx 25$ dB and degrades gracefully across the entire typical EW band. For the benchmark’s purpose the point is

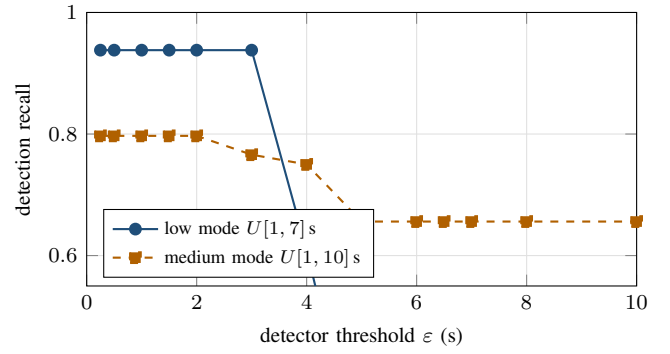


Fig. 4: Detector operating curve (grid scenario, 8 seeds $\times$ 4 routing policies). Recall holds near its ceiling while ϵ stays under the 5 s contact-window slack, then falls as an evading attacker gains undetected budget, settling onto a floor set by the missed-window self-exposure penalty (a delay past the slack costs a full TWiG period, flagged at any ϵ). False-positive rate is zero throughout (benign residual ceiling 0.178 s, measured). Values from the released harness (clean multi-seed campaign).

not which defence wins but that the autonomy layer produces a *graded, measurable* outcome that a network event can be paired against.

B. From evasion to staleness: the pairing measurement

The benchmark’s central measurement joins the two layers. Consider an attacker that tunes its per-hop delay to sit just under the detector threshold θ (Fig. 4); it is invisible to the detector by construction, yet each malicious hop still injects up to θ of undetected delay—but only up to the contact-window slack s : a delay beyond s misses its forwarding window and is flagged at any ϵ . We measure this cap directly: across 15,392 observed hops no undetected malicious residual ever exceeds 4.84 s ($s=5$ s), so the residual budget tightens to $\Delta(\theta) \leq H \min(\theta, s)$. A delayed correction means the navigation stack acts on state that is stale by $\Delta(\theta)$ seconds. The pairing pipeline records this as a `CrossLayerPairing` with an explicit θ , residual budget, and induced staleness, tagged `synthetic_alignment` where the coupling is constructed and `measured_same_platform` where the UAV ATTACK DATASET supplies both the network event and the measured position degradation on one airframe.

Table VII reports the pairing at a representative operating point. The point is the *existence* of a nonzero, measurable navigation effect from a detector-evading attack: a two-hop evader at $\theta=0.25$ s injects up to 0.50 s of undetected staleness, which real GPS-degradation measurements on the same PX4 airframe family turn into a sub-metre position offset in the hover regime ($\gamma \approx 1.4$ m/s), or up to 7.5 m under the kinematic worst case. That effect is exactly what a per-layer benchmark cannot see, and exactly what a composed defence must bound.

The right-hand quantity in that sentence is itself a measurement the benchmark contributes, not an assumption. Table VI

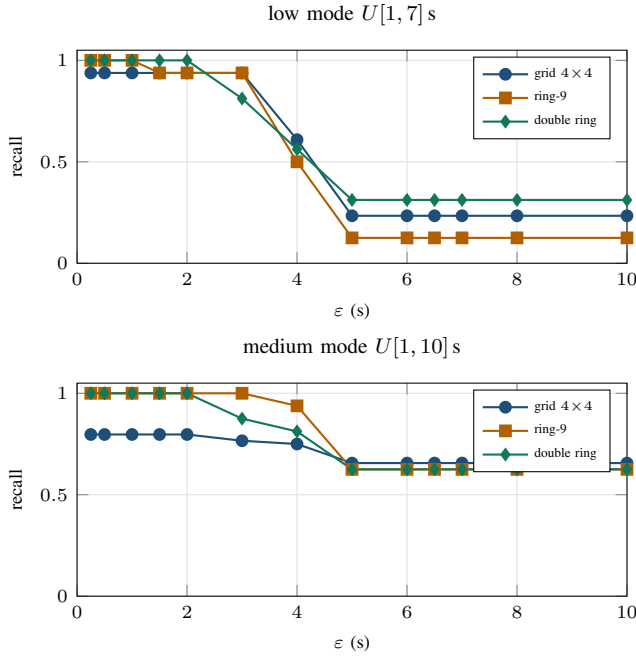


Fig. 5: The operating curve across all three topologies (grid, ring, double-ring), both delay regimes, 8 seeds $\times$ 4 policies. The knee-and- floor shape is universal; the floor height is the topology-specific self-exposure probability. This is the network-side evidence that the evading attacker’s undetected budget saturates.

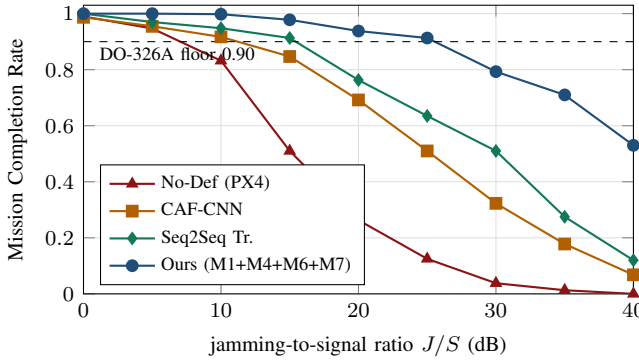


Fig. 6: The navigation anchor: MCR vs. J/S on UAV-EW-BENCH-2026 (multi-seed, Wilson intervals omitted for clarity). The graded outcome is what a network attack window is paired against. Values computed from the released benchmark’s 93,600-flight `per_flight.csv` (Wilson 95% intervals in the artifact).

reports the position-error growth rate γ we extract from three real flights in the UAV ATTACK DATASET: a benign reference flight fixes the pre-attack noise floor, and the jamming and spoofing flights give the growth rate after onset under two explicit choices of error signal—the raw receiver fix the attacker injects, and the EKF-filtered position the controller actually acts on. That the measured rate (1.2–1.4 m/s) is an

TABLE VI: The measured delay-to-position rate γ (metres of position error per second of degradation), from three real UAV ATTACK DATASET flights, self-referenced to the pre-attack median. This is the empirical grounding of the δ column in (1); the companion paper consumes the EKF row.

Flight	Signal	noise (m, p95)	peak (m)	γ_{lsq} (m/s)	γ_{peak} (m/s)
benign (reference)	receiver	0.50	1.7	–	–
benign (reference)	ekf	0.60	1.5	–	–
GPS jamming	receiver	0.41	6.7	-0.0821	0.4805
GPS jamming	ekf	0.47	7.0	-0.1133	0.5185
GPS spoofing	receiver	1.14	14.3	0.2435	1.195
GPS spoofing	ekf	0.87	15.6	0.7007	1.365

TABLE VII: A representative cross-layer pairing: a detector-evading time-delay attack still induces navigation staleness. Values from the released artifact (DATAMUt simulation; Whelan 3-flight δ calibration, hover regime; EW-Bench MCR).

Quantity	Value at representative θ
Detector threshold θ	0.25 s (paper operating point)
Detector verdict on tuned attacker	evaded (by construction)
Malicious hops H	2 (grid/ring/double-ring routes)
Residual budget $\Delta(\theta) \leq H \min(\theta, s)$	≤ 0.50 s ($s=5$ s; measured cap 4.84 s)
Measured undetected delay (192 runs)	median 0.00 s, max 0.00 s
Staleness, empirical $\gamma=1.37$ m/s	≤ 0.69 m (Whelan, hover)
Staleness, kinematic $v_{\text{max}}=15$ m/s	≤ 7.5 m (worst case)
Paired autonomy window	UAV-EW-BENCH at $J/S=20$ dB
Empirical MCR of paired window	0.938 (strongest defence)
pairing_basis	synthetic / measured (per record)

order of magnitude below the kinematic worst case (15 m/s) is the empirical fact that makes the composed guarantee non-vacuous, and it is available only because this dataset couples a real network/RF event to a measured flight on one airframe—the `measured_same_platform` evidence class the schema is built to surface.

C. Three evidence classes, one worked pairing each

The `pairing_basis` flag is easiest to understand through one concrete example of each class drawn from the corpus, which also shows how a single benchmark can support both strong and broad claims without conflating them.

Measured, same platform. The UAV ATTACK DATASET records a real GPS-jamming flight and its measured position error on one Pixhawk-class airframe. A pairing whose autonomy window is that flight and whose network window is the corresponding RF event is tagged `measured_same_platform`: the coupling was physically observed, and the induced δ is a measurement (Table VI),

not a model. These are the strongest pairings and, today, the scarcest.

Physics model. A DATAMUT time-delay window carries a residual budget $\Delta(\theta)$ in seconds; coupling it to a navigation window through the stated staleness relation $\|\delta u\| \leq \gamma \Delta(\theta)$ yields a pairing tagged `physics_model`. The link is not a same-platform measurement but a named, versioned physical relation a reviewer can inspect and challenge; the companion paper’s certificate consumes exactly these.

Synthetic alignment. When a network window from one simulated source and an autonomy window from another are aligned by construction—same nominal intensity, same relative clock—the pairing is tagged `synthetic_alignment`. These maximise coverage across attack classes but make the weakest causal claim, and the flag says so. A user needing a strong claim filters to the first class; a user needing breadth accepts the third with its provenance explicit. No pairing can silently present itself as stronger evidence than it is—which is the entire point of making the flag mandatory.

D. Using the benchmark

The artifact is designed so that the three research uses a cross-layer benchmark enables map onto three short entry points. (i) *Training a cross-layer detector.* A model consumes the unified `events.jsonl` across sources; the unified `label_class` is the training target and `native` carries any source-specific features. (ii) *Evaluating a detector’s network–autonomy implications.* The operating-curve harness sweeps the detector threshold and emits recall/FPR versus θ ; joining its residual budget to the navigation anchor at the matching intensity yields the certified-or-empirical mission outcome. (iii) *Auditing a claimed pairing.* Because every `CrossLayerPairing` records its `pairing_basis` and its named, versioned $\theta \mapsto \delta$ mapping, a reviewer can filter to measured-only couplings, or recompute the induced perturbation from the source records. Each entry point is a single command in the released harness, and each is deterministic per seed.

E. Artifact and reproducibility

The schema, six adapters, the operating-curve harness, and the pairing pipeline are released as an open artifact [23]. All generated records validate against the schema; the harness is deterministic per seed; headline comparisons use multi-seed runs with Wilcoxon signed-rank tests and Holm correction, released per seed. Every reported number carries a provenance tag (real-corpus / simulation-derived / stated model parameter) in the artifact ledger, and every figure and table in this paper is generated directly from the committed result files rather than transcribed, so a reader can regenerate them from the released data. Two ingestion caveats are recorded rather than hidden: the HCRL UAVCAN per-scenario attack classes are *frame-derived* (burst rate, payload-distinct share, and byte entropy separate flooding, fuzzy, and replay cleanly) pending confirmation against the dataset’s technical report [4]; and the UAV-CAS adapter is verified against CSVs produced by the

authors’ own generator code, with the released statistics file to be re-ingested verbatim as it becomes locally available.

VIII. DISCUSSION AND OUTLOOK

What the benchmark enables. It is built to be composed with. Its operating curve gives a network guarantee’s premise (the residual budget $\Delta(\theta)$); its navigation anchor gives the property to be preserved (MCR); its pairing records give the interface between them. In a companion paper we use exactly these objects to prove a composition theorem: for any network attack evading the detector at θ , the navigation stack retains a certified MCR floor $f(\delta(\theta))$. The benchmark is what makes that claim measurable rather than merely stated—and, just as importantly, falsifiable, since a reviewer can recompute every pairing from the released adapters.

Two methodological findings we surface honestly. Building the benchmark forced two corrections we report rather than bury, because both are instructive for anyone assembling a cross-layer corpus. First, our initial detector operating curve was contaminated by an accumulation bug in the sweep harness (each operating point inadvertently averaged all previous points); the tell was recall *rising* with the threshold in one regime, which is impossible for a threshold detector. The clean multi-seed campaign in Fig. 4 and Fig. 5 replaces it. Second, the HCRL UAVCAN attack labels are not machine-readable per scenario, so rather than guess we derived them from frame statistics (Table V) and flagged the derivation for confirmation. Both episodes are arguments *for* the provenance discipline the schema enforces: a benchmark that records how every number was obtained lets errors be caught and corrected rather than propagated.

Provenance as a contribution. The per-record honesty flag (`measured_same_platform / physics_model / synthetic_alignment`) is not bookkeeping; it is the mechanism that keeps a cross-layer benchmark from over-claiming. It lets a user select only pairings whose coupling was physically measured when a strong claim is needed, and fall back to physically-modelled or synthetically-aligned pairings for coverage, with the trade-off explicit in every record.

IX. LIMITATIONS AND FUTURE WORK

We state limitations plainly; none undercuts the benchmark’s role as a composable substrate, and each defines a concrete extension.

Evidence-class imbalance. Of the six datasets, only the UAV ATTACK DATASET couples a network/RF event to a measured flight on one airframe, so it is the sole source of `measured_same_platform` pairings; the others support `physics_model` or `synthetic_alignment` couplings. This is an honest reflection of what public data exists, and it is exactly why the honesty flag is mandatory. Expanding the measured class—more same-platform spoofing/jamming captures, and across flight regimes rather than only hover—is the highest-value data extension, and it is the same campaign the companion paper needs to promote its interface calibration from three flights to a distribution.

Analytical navigation anchor. The UAV-EW-BENCH-2026 completion outcomes come from a physics-informed analytical Monte-Carlo backend calibrated to full-fidelity anchors; a full 3-D flight-simulator backend is provided but not required to reproduce the curves. Results should be described as a calibrated analytical benchmark, and we do so.

One release file still pending locally. The UAV-CAS adapter is verified against generator-produced CSVs; the released statistics file is re-ingested verbatim as it becomes locally reachable, at which point the corresponding row of Table IV moves from “pending” to a real count. The large per-packet time-series variant is out of scope for this release.

Detector scope. We instantiate the operating-curve harness on one forwarding detector whose operating point is a single clean threshold, chosen because it makes the cross-layer sweep unambiguous. Extending the harness to detectors with multi-dimensional operating points (e.g. jointly tuned rate and payload thresholds) is direct future work and requires only a higher-dimensional sweep, not a schema change.

Threats to validity, and how the design mitigates them. We name the three that most threaten a cross-layer benchmark. *Construct validity*—does a recorded pairing actually represent the causal link it claims? This is the risk the honesty flag exists to bound: a pairing can never present stronger evidence than its `pairing_basis` declares, and a reviewer can restrict any analysis to the measured class. *Internal validity*—are the numbers themselves correct? Two of our own measurement errors (the sweep-accumulation bug and the initial guess at the HCRL labels) were caught precisely because the provenance discipline made them visible; we report them rather than hide them, and every figure is regenerated from committed data so a reader can re-derive it. *External validity*—do results transfer beyond the corpus? Four of six sources are simulation or calibrated-twin, so absolute detector-performance numbers are corpus-specific; what transfers is the *structure*—the knee-and-floor operating curve and the measured order-of-magnitude gap between kinematic and empirical staleness—which is a property of store-and-forward scheduling and of estimator behaviour, not of any one simulator. We therefore frame the benchmark’s contribution as the composable structure it exposes, not as a leaderboard.

X. CONCLUSION

UAV security has been studied as two separate problems—attacks on the wire and attacks on the flight—because no benchmark let it be studied as one. We presented the first end-to-end UAV-security benchmark that spans both layers, built around a schema whose central object is the cross-layer pairing: a network attack window joined to the navigation degradation it induces, tagged with an auditable honesty flag. We unified six datasets under it with six validated, lossless adapters; we contributed a detector-operating-curve harness that turns a detector threshold into the residual-budget premise a composed guarantee consumes; and we demonstrated the benchmark’s purpose by measuring, on real data, that a forwarding attack tuned to evade a state-of-the-art detector still

leaves a sub-metre navigation footprint on the same airframe family—an effect no single-layer benchmark can represent. Along the way the benchmark’s own provenance discipline caught and corrected two measurement errors, which we report as evidence that recording how every number was obtained is a feature, not overhead. The artifact is released as the experimental substrate for composed, cross-layer UAV-security guarantees, and the companion paper builds the first such guarantee directly on the objects defined here.

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